

KONOLLA SHANMUKA SAI

Project Engineer | QA Engineer | Design Engineer

 konollashanmukasai@gmail.com  (+91) 9133087960

 H No.: 4-39, DVK Road, Haliya, Telangana State-508377, India

OBJECTIVE

Results-driven Mechanical Engineer with 3+ years of aerospace quality assurance, and defence project integration. Proven expertise in GD&T, NDT evaluation, composite material testing, QAP standardization, and site integration activities in DRDO-affiliated laboratories. Adept at ensuring compliance with aerospace standards through meticulous inspection, documentation, and process control..

WORK EXPERIENCE

09/2024 – Present	Rail Based Mobile Launcher, Project Engineer - Supported mechanical integration of subsystems and assemblies ensuring compliance with design requirements. - Coordinated with design, manufacturing, quality, and integration teams for smooth fitment and execution. - Applied QA expertise to maintain inspection standards and dimensional accuracy during integration. - Participated in site inspections throughout system assembly and integration phases. - Collaborated with cross-functional teams across project locations for effective execution. - Involved in Endurance Testing and Environmental Stress Screening (ESS) for system validation. - Performed GD&T validation during inspection and acceptance stages of launcher components. - Ensured compliance of components with QAPs, ISO/AS standards, and customer specifications. - Conducted root cause analysis for system- and component-level issues, supporting corrective actions. - Supported integration, inspection, and testing of composite structures, verifying fitment, bonding integrity, and dimensional conformity. - Worked closely with Quality Engineering teams to ensure compliance with ISO/AS standards, inspection plans, and customer specifications.
03/2023 – 09/2024	DR&QA, DRDL, Quality Assurance Engineer - Conducted vendor, in-process, and on-site inspections for aerospace and defence hardware ensuring compliance with approved standards. - Prepared and reviewed Quality Assurance Plans aligned with MIL, ASTM, and IS standards. - Applied GD&T principles for dimensional and geometric verification of aerospace components. - Validated engineering drawings against specifications and applicable standards to ensure design integrity. - Coordinated and verified UT, RT, C-scan, and Phased Array UT for metallic and composite parts. - Participated in destructive testing for qualification and validation of aerospace systems. - Reviewed and evaluated vendor test, inspection, and qualification reports for compliance.

- Performed root cause analysis for inspection failures and test deviations, supporting corrective actions.
- Inspected composite lay-up, ply orientation, curing, bonding, and surface defects such as delamination and porosity.
- Witnessed Tensile, flexural, and ILSS tests of composite specimens per ASTM standards.
- Reviewed and approved Material Test Reports for prepreg, adhesives, and resin systems ensuring traceability and shelf-life compliance.
- Supported failure analysis of composite components, documenting root causes and corrective measures

09/2022 – 03/2023

DYSL, DRDO, Project Engineer

- Worked as a Project Engineer at DYSL, focusing on smart materials and their applications in aerospace systems
- Involved in design and development of miniature models for aerospace applications
- Supported concept design, modelling, and integration of smart material-based components
- Assisted in mechanical design and layout of miniature assemblies for functional validation
- Evaluated material behaviour, performance, and feasibility for aerospace usage
- Coordinated with R&D, testing, and integration teams during project execution
- Supported testing and performance evaluation of smart material-based models
- Simulation of smart material-based actuators and fabrication of components.

EDUCATION

2018 – 2022

Bachelor of Engineering, Mechanical Engineering,

Maturi Venkata Subba Rao (M.V.S.R.) Engineering College

- CGPA: 7.47/10

2016 – 2018

Hyderabad, India

Intermediate, MPC, Narayana junior college, Hyderabad

- Percentage: 95%

2016

Halia, India

Matriculation, VSR High school, Halia

- CGPA: 8.2/10

SKILLS & INTERESTS

Office software

MS Office (Word, Excel and PowerPoint)

Software

- AutoCAD
- SolidWorks
- Basics of ANSYS

Languages known

- English, Telugu and Hindi

INTERNSHIPS

Non-Destructive testing Intern, National Academy for Training & Technical Services (NATTS)

- Gained knowledge in inspection, testing, or evaluating materials, components or assemblies for discontinuities, or differences in characteristics without destroying the work piece.

STRENGTHS

Core Strengths

- Communication skills
- Problem-solving skills
- Time management skills
- Interpersonal skills
- Adaptability